# EXPERIMENT 12

## Implementing Security Features and Governance / Administration Practices in Zoho Creator

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## Aim

To implement and explore the security, access-control, data-sharing and governance features of a Zoho Creator application.

## Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

## Application Used

The practical was carried out in Zoho Creator using the Student Admission Enquiry Form application. The application is used for managing student admission enquiries, and it served as the practical environment for exploring the security and governance features described in this record. The Zoho Creator account was on a trial plan, so some advanced features may require an upgrade; no upgrade-only feature is claimed as enabled or configured in this record.

## Security and Governance Architecture

Figure 1 is a conceptual supporting diagram of the security and governance structure of the Student Admission Enquiry Form application. It is not a screenshot of the Zoho Creator interface.

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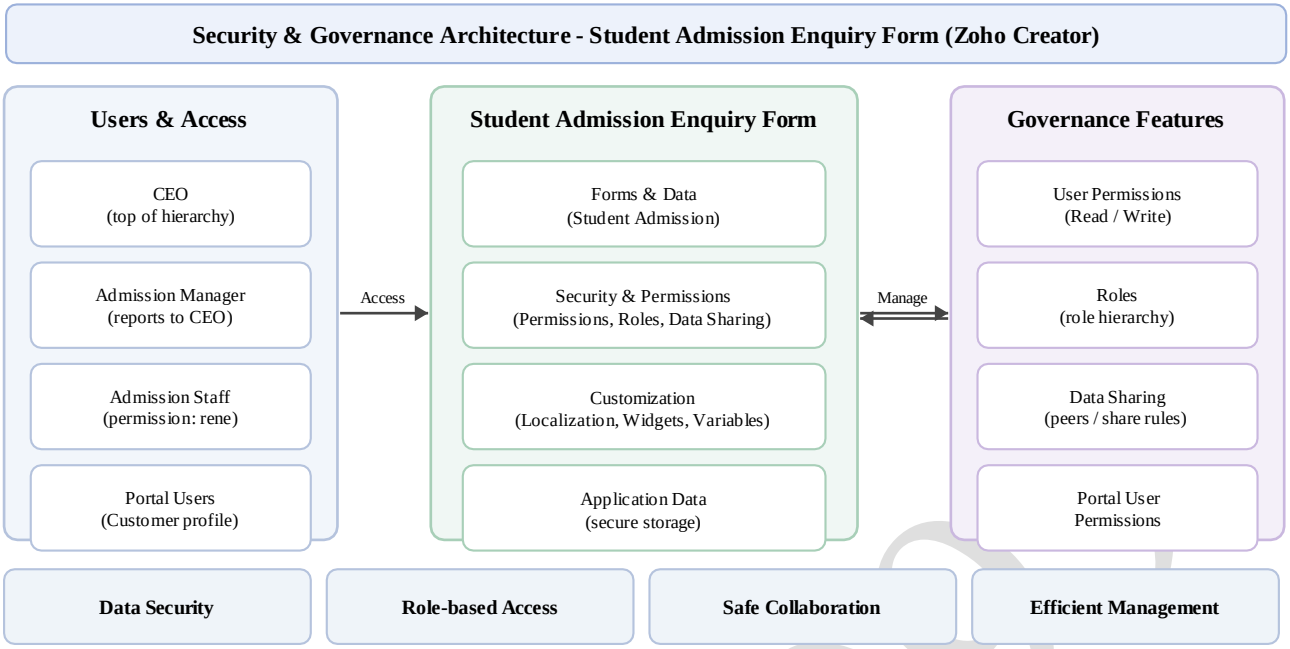



***Figure 1: Security & Governance Architecture (Conceptual Diagram)***

The diagram shows users (such as the CEO, Admission Manager, Admission Staff and portal users) accessing the application, which contains forms and data, security and permissions (user permissions, roles and data sharing) and customisation items. The governance features on the right (user permissions, roles, data sharing and portal user permissions) manage and control access to the application. The key outcomes are data security through controlled access, role-based access, safe collaboration through data sharing, and efficient management.

## User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions tab lists the permission sets available in the application and provides an Add Permission option. The application contains two permission profiles:

| Name | Description (as displayed) |
|---|---|
| rene | This profile will have read permission for all components |
| Write | This profile will have write permission for all components |

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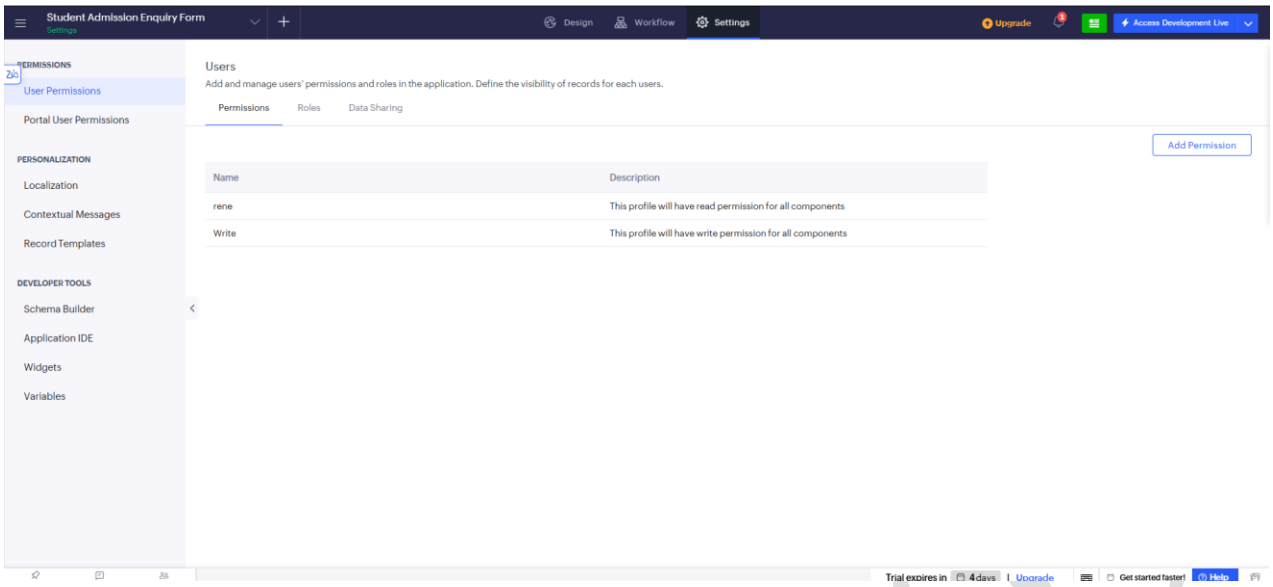



***Figure 2: User Permissions in the Student Admission Enquiry Form application***

Read permission allows a user to view the components and records of the application without changing them, whereas Write permission allows the user to make changes, such as adding or modifying records. The custom profile "rene" was created with read permission, alongside the built-in Write profile.

## Role-Based Access Control

The Roles tab shows the role hierarchy of the application. In the application, the role hierarchy is CEO at the top, Admission Manager below the CEO, and Admission Staff below the Admission Manager. The options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available.

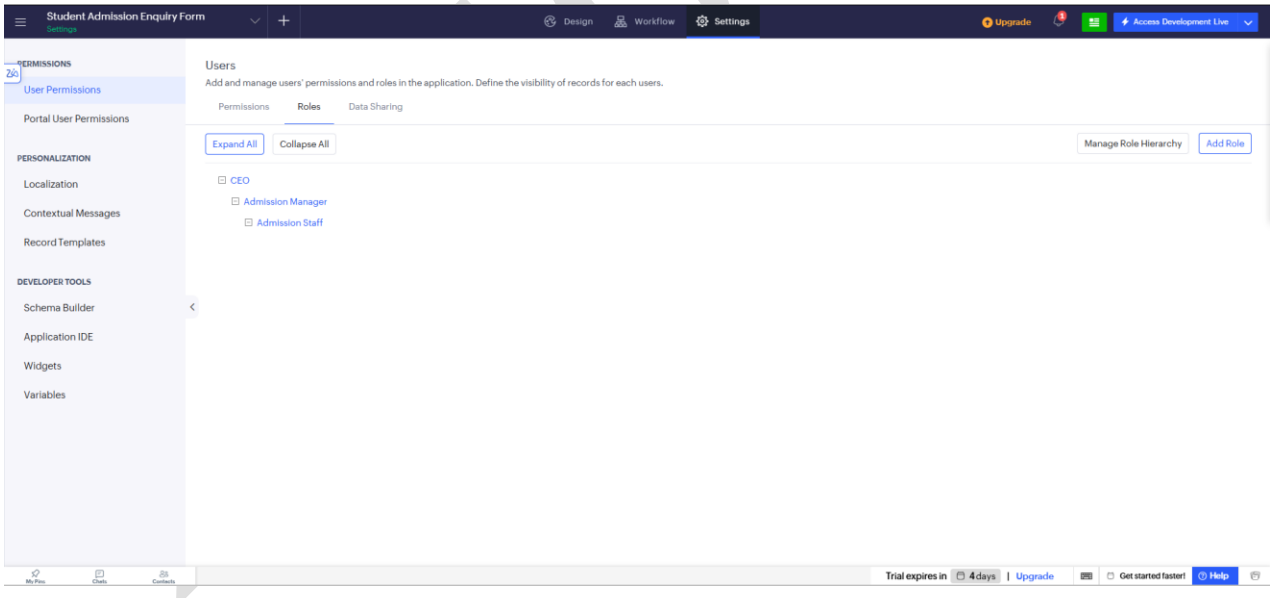



***Figure 3: Roles and Role Hierarchy in the Student Admission Enquiry Form application***

Selecting the Admission Manager role displays its details, as follows:

| Field | Value |
|---|---|
| Role Name | Admission Manager |
| Reports To | CEO |
| Description | Manages student admission enquiries |
| Share Data with Peers | Yes |
| Associated Users | None at the time of the screenshot |

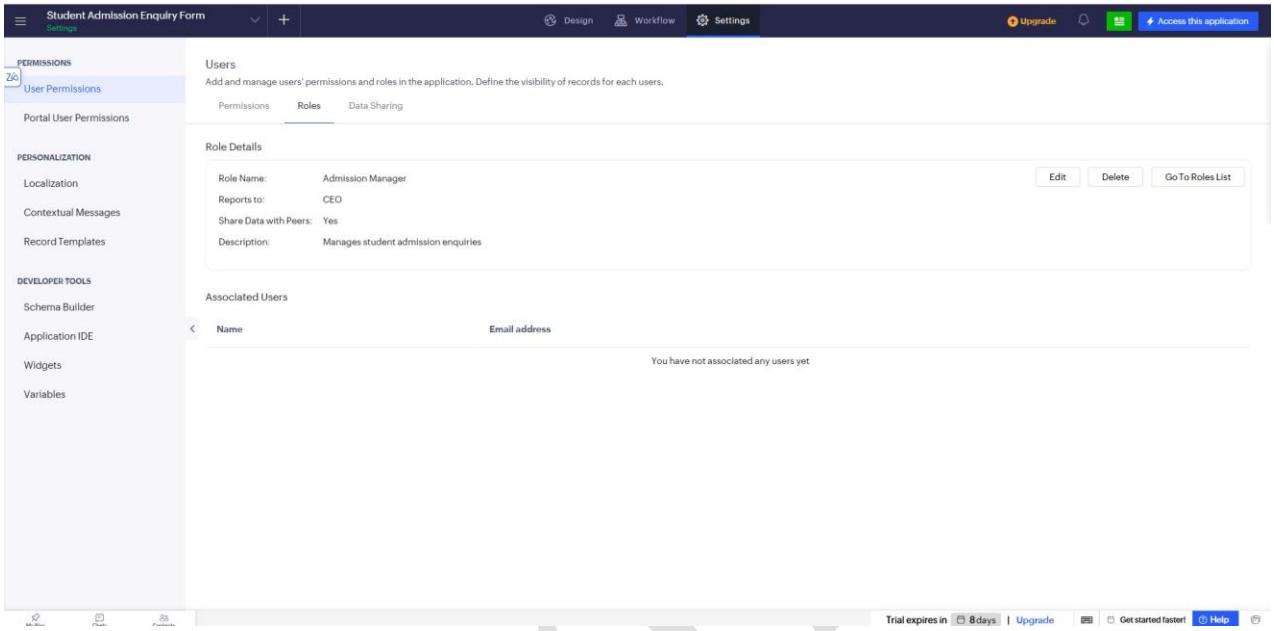



***Figure 4: Role Details – Admission Manager role with its reporting hierarchy***

The "Reports To" field builds the reporting chain (Admission Manager to CEO) that Zoho Creator uses to decide which records a role can see above its own. The Associated Users table shows who holds the role; at the time of the screenshot, no users were associated with the Admission Manager role.

## Data Sharing

Data sharing controls the visibility of the records of a form beyond the default visibility of the role hierarchy. A share rule, created from the Data Sharing tab, specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this application, visibility among users of the same role is extended through the "Share Data with Peers" setting of the Admission Manager role (Figure 4), which is set to Yes. As a result, users placed in the Admission Manager role can view each other's admission enquiry records. No separate share rule is recorded in this experiment.

## Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users, such as applicants who interact with the application from outside the organisation. The Permissions tab shows the "Customer" profile, described as a profile with add and view permission. This allows an applicant to submit a student admission enquiry and view it, without being able to change or delete other records.

## Security Features and Governance Controls

Figure 5 is a conceptual academic diagram that groups the security and governance controls of Zoho Creator. It is not a screenshot, and it is not proof that every listed feature was configured.

**Security Features and Governance Controls in Zoho Creator**




***Figure 5: Security Features and Governance Controls in Zoho Creator (Conceptual Diagram)***

| Control | Explanation |
|---|---|
| User access control | User permissions (Read and Write), role-based access and portal user permissions determine who can use the application and what they can do. |
| Data governance | Data sharing, read / write access, record visibility and role hierarchy govern how application data is accessed. |
| Application governance | Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application. |
| Role hierarchy | Arranges roles in a reporting structure that influences which records a role can see. |
| Data sharing | Extends the visibility of records from one role or user to another with a chosen access level. |
| Record visibility | Decides which records are visible to which users, based on roles and sharing rules. |
| Administrative controls | Allow an administrator to manage permissions, roles and application settings. |

In this practical, user permissions, the role hierarchy, the Admission Manager role details and portal user permissions were examined. The other items in the diagram, such as localization, schema builder and widgets, are available in the Settings menu of the application, but they were not configured in this experiment.

## Role-Based Access and Data Sharing Flow

Figure 6 is a conceptual diagram of the security flow: User → User Permissions → Role → Data Sharing → Access Level → Student Admission Data. It represents how access is controlled in principle and is not evidence that each step was configured.

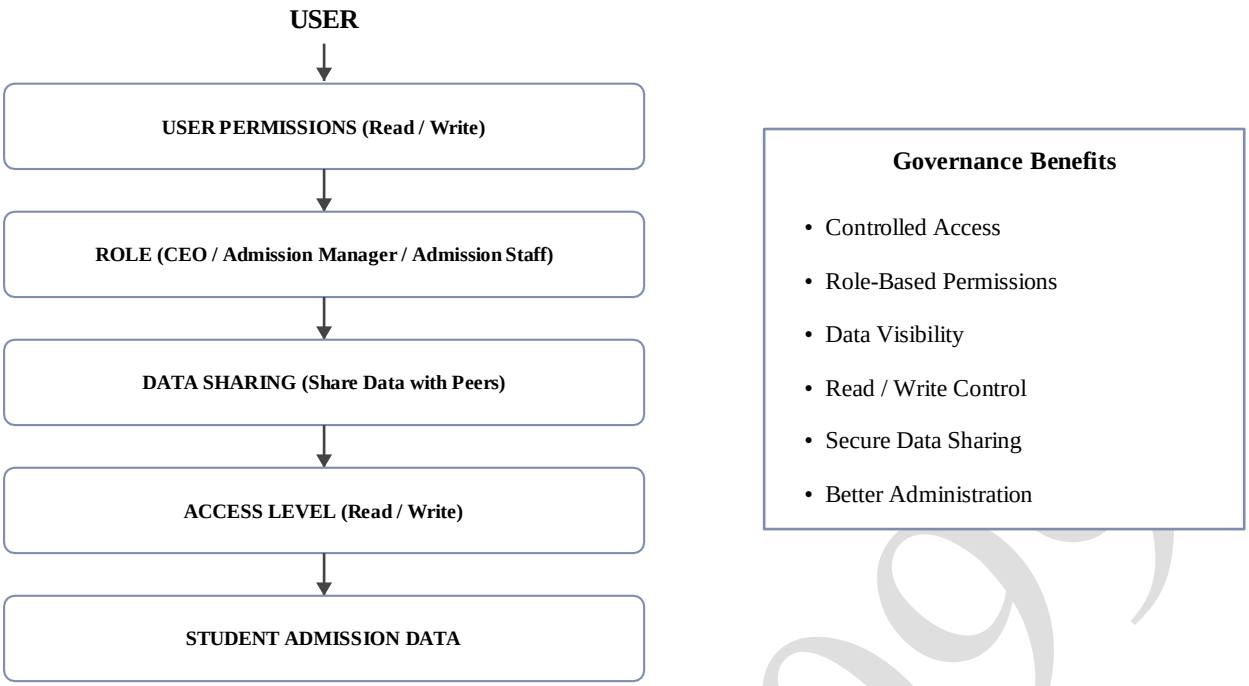



***Figure 6: Role-Based Access and Data Sharing Flow (Conceptual Diagram)***

- A user is first given permissions (Read / Write).
- The user belongs to a role, such as CEO, Admission Manager or Admission Staff, which determines the position in the hierarchy.
- Data sharing, such as Share Data with Peers, extends the visibility of records among users.
- The access level (Read / Write) defines what the recipient can do, and this controls access to the Student Admission data.

## Governance Observations

- Read-level (rene) and Write permission profiles were available in the application.
- The role hierarchy CEO → Admission Manager → Admission Staff was viewed, and the Admission Manager role is set to report to the CEO.
- The Admission Manager role has Share Data with Peers set to Yes.
- Portal User Permissions showed a Customer profile with add and view permission.

## Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the rene (read) and Write permission profiles were examined, the role hierarchy of CEO, Admission Manager and Admission Staff was viewed, and the Admission Manager role was set to report to the CEO with Share Data with Peers enabled for the Student Admission Enquiry Form application.

Portal user permissions restrict external applicants to a Customer profile with add and view permission. The experiment therefore shows the access-control tools that Zoho Creator offers. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications.

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